

The effect of obesity on polycystic ovary syndrome: a systematic review and meta-analysis.

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While many women with polycystic ovary syndrome (PCOS) are overweight, obese or centrally obese, the effect of excess weight on the outcomes of PCOS is inconsistent. The review aimed to assess the effects of overweight, obesity and central obesity on the reproductive, metabolic and psychological features of PCOS. MEDLINE, EMBASE, CINAHL, Cochrane Central Register of Controlled Trials (CENTRAL) and PSYCINFO were searched for studies reporting outcomes according to body mass index categories or body fat distribution. Data were presented as mean difference or risk ratio (95% confidence interval). This review included 30 eligible studies. Overweight or obese women with PCOS had decreased sex hormone-binding globulin (SHBG), increased total testosterone, free androgen index, hirsutism, fasting glucose, fasting insulin, homeostatic model assessment-insulin resistance index and worsened lipid profile. Obesity significantly worsened all metabolic and reproductive outcomes measured except for hirsutism when compared to normal weight women with PCOS. Overweight women had no differences in total testosterone, hirsutism, total-cholesterol and low-density lipoprotein-cholesterol compared to normal weight women and no differences in SHBG and total testosterone compared to obese women. Central obesity was associated with higher fasting insulin levels. These results suggest that prevention and treatment of obesity is important for the management of PCOS. © 2012 The Authors. obesity reviews © 2012 International Association for the Study of Obesity.